

TED (15)-5072

REVISION-2015

MODEL QUESTION PAPER

FIFTH SEMESTER DIPLOMA EXAMINATION IN CHEMICAL ENGINEERING

ORGANIC TECHNOLOGY

Maximum marks: 100

Time: 3 hrs

Marks

		PART-A (Maximum marks: 10)	
		Answer the following questions in one or two sentences. Each question carries two marks	
I	1)	Define Iodine value of oil.	2
	2)	Define rancidity of oil.	2
	3)	List the important chemicals used for bleaching pulp.	2
	4)	Define permissibility of an explosive.	2
	5)	List the important operations used in sugar refining.	2
		PART-B (Maximum marks: 30)	
II		Answer any five of the following question. Each question carries 6 marks	
	1	Differentiate saturated and unsaturated fatty acids with examples.	6
	2	State the purpose of hydrogenation of vegetable oil and list the important catalysts used for the process.	6
	3	Describe the preparation of pulp stock for paper manufacture.	6
	4	Summarize the different pulp bleaching methods.	6
	5	Differentiate the properties of primary and secondary explosives.	6
	6	Explain the manufacture of dynamite.	6
	7	Describe purification of cane juice for evaporation	6
		PART-C (Maximum marks: 60) (Answer one full question from each unit. Each question carries 15 marks)	
		MODULE-I	
III	a)	Explain the extraction process of oil by expellers with a flow diagram.	8
	b)	Describe refining of crude glycerin.	7
		OR	
IV	a)	Explain the manufacture of soap by full boiled process with a flow diagram	9
	b)	Differentiate anionic, cationic and non- ionic detergents.	6
		MODULE-II	
V	a)	Explain the function of additives mixed with pulp in paper manufacture.	6

	b)	Describe the working of Fourdiner paper machine with a line diagram.	9
		OR	
VI	a)	Explain sulfate process for the manufacture of pulp with a flow diagram	9
	b)	Describe the manufacture of laminated paper and state its advantages	6
		MODULE-III	
VII	a)	Describe the manufacture of smokeless powder with a flow diagram	8
	b)	Explain the manufacture of RDX with a neat sketch	7
		OR	
VIII	a)	Explain the manufacture of Nitroglycerine with a flow diagram	8
	b)	Describe the manufacture of Bio-insecticides.	7
		MODULE-IV	
IX	a)	Describe vegetable tanning and chrome tanning processes used in leather manufacture	8
	b)	Describe evaporation and crystallization of cane juice for the manufacture of sugar.	7
		OR	
X	a)	Describe the manufacture of starch from corn with a flow diagram	8
	b)	Explain the post tanning operations to finish leather.	7